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DPAEPARTMENT	ELECTRICAL ENG (power)
SECTION	“A”
SEMISTER	2nd
SUBJECT	COMPUTER PROGRAMING

1. ***Objective***

The primary goal of this lab is to master selection and flow control in C++ using the switch statement and its associated keywords. Students will learn to:

- Implement switch as an alternative to multiple if-else blocks.
- Utilize break to manage program flow and prevent "fall-through".
- Apply the continue statement within loops.
- Develop interactive, menu-driven programs.

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2. ***Technical Overview***

The Switch Statement

The switch statement is a **multiple-selection statement** that executes a specific block of code based on the value of a constant integral expression (integers or characters).

Key Components:

- **Case:**

  Defines the starting point for execution if the expression matches the case value.

- **Break:**

Terminstes the statement sequence and jumps the control flow to the line following the switch block.

- ***Default:***

  An optional section that executes if no case matches the provided expression.

Important Rules

- The switch expression must result in a **constant value**.
- **Duplicate case values** are strictly prohibited.
- If break is omitted, execution will continue into the next case automatically.

EXAMPLE # 1

```
#include <iostream>
using namespace std;
int main()
    {

        char grade=0;
        cout<<"enter your grade (A,B,C,D,E,F)"<<endl;
        cin>>grade;

        switch(grade)
    {
            case 'A':
            case 'a':
cout<<"excellent";
break;


        case'B':
            case 'b':
cout<<"very good";
break;

    case 'C':
        case 'c':
cout<<"good";
break;

    case 'D':
        case 'd':
cout<<"Pass";
break;

    case 'E':
        case 'e':
cout<<"barely passed";
break;

    case 'F':
        case 'f':
cout<<"repeat again";
break;


default:
    cout<<"enter correct grade";

}

return 0;
}
```

EXAMPLE # 2

```
#include <iostream>
using namespace std;
int main()
    {

    int a=0;
    cout<<"select one option\n 1.convert kilometers to meters\n 2.convert meters to centimeters\n 3convert kilogram to gram\n 4.convert celcius to fahrenheit\n";
    cin>>a;

    switch (a){

        case 1:{

        float kilometers=0;
        float meters=0;
        cout<<"enter kilometers"<<endl;
        cin>>kilometers;
        meters=kilometers*1000;
        cout<<"there are  "<< meters<<"  meters in  "<<kilometers<<"  kilometers  ";
        break;
}
        case (2):{
        float meters=0;
        float centimeters=0;
        cout<<"enter meters"<<endl;
        cin>>meters;
        centimeters=meters*100;
        cout<<"there are  "<< centimeters<<"  centimeters in  "<<meters<<"  meters  ";
        break;
    }

    case (3):{
        float kilogram=0;
        float grams=0;
        cout<<"enter kilograms"<<endl;
        cin>>kilogram;
        grams=kilogram*1000;
        cout<<"there are  "<< grams<<"  grams in  "<<kilogram<<"  kilograms  ";
        break;
    }

    case (4):{
        float celcius=0;
        float fahrenheit=0;
        cout<<"enter celcius"<<endl;
        cin>>celcius;
        fahrenheit=(celcius*9/5) + 32;
        cout<<"there are  "<< fahrenheit<<"   fahrenheit in  "<<celcius<<"  celcius  ";
        break;



    }
}

return 0;
}
```

EXAMPLE # 3

```
#include <iostream>
using namespace std;
int main()
    {
        int day=0;
        cout<<"enter a number ( 1 - 7)"<<endl;
        cin>>day;

        switch(day)
    {
            case (1):
cout<<"monday";
break;


        case(2):
cout<<"tuesday";
break;

    case(3):
cout<<"wednesday";
break;

    case(4):
cout<<"thursday";
break;

    case(5):
cout<<"friday";
break;

    case(6):
cout<<"saturday";
break;

    case(7):
cout<<"sunday";
break;

default:
    cout<<"enter a valid number";

}

return 0;
}
```